

Lab 4 plus (TAN ZHI MING A23CS0189)

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

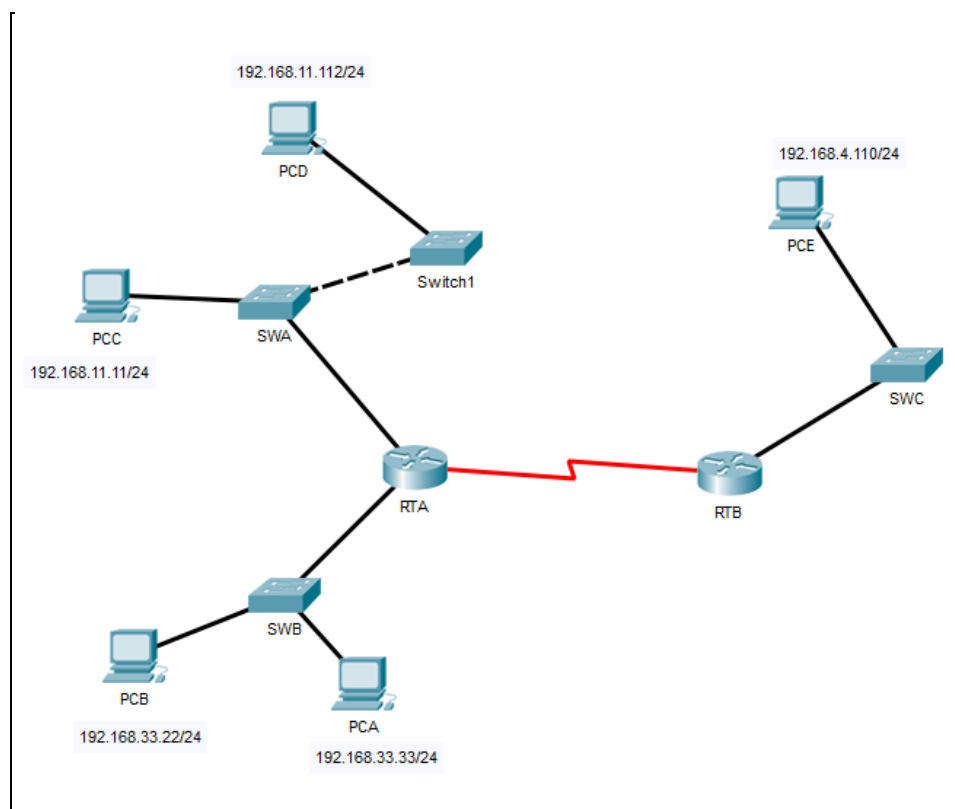


Figure 1

Part 1: Review the topology

Step 1: Open the *Lab_4_ARP.pkt*, and Perform the following tasks.

- a. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol Address      Age (min)  Hardware Addr  Type   Interface
Internet 192.168.11.1      -         0002.4A00.0E91  ARPA   FastEthernet1/0
Internet 192.168.33.1      -         000C.CF0C.593A  ARPA   FastEthernet0/0
```

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
Protocol Address      Age (min)  Hardware Addr  Type   Interface
Internet 192.168.4.1      -         0001.977A.B614  ARPA   FastEthernet0/0
```

- c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp
SWA#show mac-address-table
```

```
SWA#show mac-address-table
      Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1     0002.4a00.0e91    DYNAMIC   Fa0/1
1     000c.8546.7d85    DYNAMIC   Fa1/1
```

SWA

```
SWB#show arp
```

```
SWB#show mac-address-table  
Mac Address Table
```

Vlan	Mac Address	Type	Ports
----	-----	-----	-----
1	000c.cf0c.593a	DYNAMIC	Fa0/1

SWB

```
SWC>enable  
SWC#show arp
```

```
SWC#show mac-address-table  
Mac Address Table
```

Vlan	Mac Address	Type	Ports
----	-----	-----	-----
1	0001.977a.b614	DYNAMIC	Fa0/1

SWC

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type **arp -a** and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.

```
Cisco Packet Tracer PC Command Line 1.0  
C:\>arp -a  
No ARP Entries Found  
C:\>
```

PCA

```
Cisco Packet Tracer PC Command Line 1.0  
C:\>arp -a  
No ARP Entries Found  
C:\>
```

PCB

```
Cisco Packet Tracer PC Command Line 1.0  
C:\>arp -a  
No ARP Entries Found  
C:\>
```

PCC

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

PCD

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
```

PCE

- e. What are your thoughts on the results?

No entry for all PCs because all of the PCs haven't receive any ARP request. Thus, there will be no entry in the PC ARP table.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- Click **PCA** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command. This may take a few seconds.
- In the Command prompt of PCA, type **arp -a**. Paste the result of this command here.

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.33.22         0060.47ea.a746        dynamic
```

- In the Command prompt of PCB, type **arp -a**. Paste the result of this command here

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.33.33         0002.1755.9a06        dynamic
```

- In the Command prompt of PCC, PCD abd PCE, type **arp -a**. Paste the result of this command here.

```
C:\>arp -a
No ARP Entries Found
```

PCC

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCD

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCE

Step 2: Generate traffic between PCC to all other PC except PCA.

- a. Click **PCC** and click the Desktop tab > Command Prompt.
- b. Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp -a**. Paste the result after these commands here.

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91       dynamic
```

- c. Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp -a**. Paste the result after these commands here.

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91       dynamic
192.168.11.112        0001.6462.0278       dynamic
```

- d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp -a**. Paste the result after these commands here.

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91       dynamic
192.168.11.112        0001.6462.0278       dynamic
```

- e. Discuss the results you got from all the commands on PCC.
After PCB,PCD and PCE received the ARP request form PCC, then only the ARP entries will shown,The ARP entries will only be shown once pinged to corresponding address.
- f. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```

RTA>enable
RTA#show arp

```

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	192.168.11.1	-	0002.4A00.0E91	ARPA	FastEthernet1/0
Internet	192.168.11.11	5	00D0.D39A.C0D9	ARPA	FastEthernet1/0
Internet	192.168.33.1	-	000C.CF0C.593A	ARPA	FastEthernet0/0
Internet	192.168.33.22	5	0060.47EA.A746	ARPA	FastEthernet0/0

```

RTA#

```

- g. At Router RTB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```

RTB>enable
RTB#show arp

```

```

RTB>enable
RTB#show arp

```

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	192.168.4.1	-	0001.977A.B614	ARPA	FastEthernet0/0
Internet	192.168.4.110	5	0060.702D.7C08	ARPA	FastEthernet0/0

```

RTB#

```

Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```

SWA>enable
SWA#show arp

SWA#show mac-address-table

```

```

SWA>enable
SWA#show arp

SWA#show mac-address-table

```

Mac Address Table			
Vlan	Mac Address	Type	Ports
1	0002.4a00.0e91	DYNAMIC	Fa0/1
1	000c.8546.7d85	DYNAMIC	Fa1/1

```

SWA#

```

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
```

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    000c.cf0c.593a    DYNAMIC Fa0/1
SWB#
```

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    0001.977a.b614    DYNAMIC Fa0/1
```

- d. Do switches use arp table? (Y/N)
Y
- e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation. It is not part of NetComm syllabi, it is just for knowledge..*

The switch has an ARP table that stores the IP and MAC addresses of network devices. The ARP table is used to determine the destination MAC address of a network node.

- f. What information does the command `show mac-address-table` gives?

Displays mac address entries on the switch. "dynamic" limits the display MAC addresses learned by the switch at runtime.

Part 3: Attach wireless lab results.

In this part, you will use **Lab_4_Wireless.pka** file.

Step1: Change the filename of the pka file.

- Change the Lab 4 filename to include your name. *Example: *Lab4AliAhmad.pkt*
- Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.

The screenshot displays the Packet Tracer interface. On the left, a network topology is shown with a switch (S1) connected to three PCs (PC1, PC2, PC3) and a wireless router (WRS2). PC1 is connected to S1 via F0/11 (172.17.10.21/24, VLAN 10). PC2 is connected to S1 via F0/18 (172.17.20.22/24, VLAN 20). PC3 is connected to WRS2 via F0/7 (172.17.40.100/24). WRS2 is connected to S1 via F0/7 (172.17.88.25/24, VLAN 88). A 'Subinterfaces' box lists: G0/0.10 172.17.10.1/24, G0/0.20 172.17.20.1/24, and G0/0.88 172.17.88.1/24. On the right, the 'Packet Tracer – Configuring Wireless LAN Access' activity window is open. It shows an 'Addressing Table' with the following data:

Device	Interface	IP Address	Subnet Mask	Default Gate
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1

At the bottom of the activity window, the 'Completion' status is shown as '100/100' with a red box highlighting the progress bar and '1/1' indicator.

- Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.

Cisco Packet Tracer - C:\Users\braya\Downloads\Lab_4_TANZHIMING.pka - Guest - 2025-01-25 22:55:19

File Edit Options View Tools Extensions Window Help

Logical Physical 05 / 72 Root 12:43:30

Subinterfaces ---
 G0/0.10 172.17.10.1/24
 G0/0.20 172.17.20.1/24
 G0/0.88 172.17.88.1/24

Part 2: Configure a Wireless Client

Step 1: Configure PC3 for wireless connectivity.
 Because SSID broadcast is disabled, you must manually configure PC3 with the correct SSID and passphrase to establish a connection with the router.

- Click PC3 > Desktop > PC Wireless.
- Click the Profiles tab.
- Click New.
- Name the new profile Wireless Access.
- On the next screen, click Advanced Setup. Then manually enter the SSID of WRS_LAN on Wireless Network Name. Click Next.
- Choose Obtain network settings automatically (DHCP) as the network settings, and then click Next.
- On Wireless Security, choose WPA2-Personal as the method of encryption and click Next.
- Enter the passphrase cisco123 and click Next.
- Click Save and then click Connect to Network.

Step 2: Verify PC3 wireless connectivity and IP addressing configuration.

- The Signal Strength and Link Quality indicators should show that you have a strong signal.
- Click More Information to see details of the connection including IP addressing information.
- Close the PC Wireless configuration window.

Part 3: Verify Connectivity
 All the PCs should have connectivity with one another.

Time Elapsed: 00:11:26 Completion: 100/100
☒ Dock

Time: 00:12:05

Scenario 0

File	Last Status	Source	Destination	Type	Color	Time(ms)	Periodic	Num	Edit	Delete
		PC3	PC2	ICMP	Orange	0.000	N	0	(edit)	(delete)
		PC3	PC1	ICMP	Green	0.000	N	1	(edit)	(delete)

Toggle PC3 List Window

Prisoner Strainight Through